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Do Vaping Prevention Messages Impact Adolescents and Young Adults? A Meta-Analysis of Experimental Studies

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ABSTRACT

Vaping prevention messages are widely used to communicate the health harms and addiction risks of vaping and discourage vaping among adolescents and young adults. We conducted a meta-analysis of experimental studies to examine the effects of these messages and to understand their theoretical mechanisms. Systematic, comprehensive searches generated 4,451 references, among which 12 studies (cumulative $N = 6{,}622$) met inclusion criteria for the meta-analysis. Across these studies, a total of 35 different vaping-related outcomes were measured, and 14 outcomes assessed in two or more independent samples were meta-analyzed. Results showed that compared to control, exposure to vaping prevention messages led to higher vaping risk perceptions, including harm perceptions ($d = 0.30$, $p < .001$), perceived likelihood of harm ($d = 0.23$, $p < .001$), perceived relative harm ($d = 0.14$, $p = .036$), addiction perceptions ($d = 0.39$, $p < .001$), perceived likelihood of addiction ($d = 0.22$, $p < .001$), and perceived relative addiction ($d = 0.33$, $p = .015$). Also, compared to control, exposure to vaping prevention messages led to more vaping knowledge ($d = 0.37$, $p < .001$), lower intentions to vape ($d = -0.09$, $p = .022$), and higher perceived message effectiveness (message perceptions; $d = 0.57$, $p < .001$; effects perceptions; $d = 0.55$, $p < .001$). Findings suggest vaping prevention messages have an impact, yet may operate through different theoretical mechanisms than cigarette pack warnings.

Vaping or use of e-cigarettes has become a significant public health issue for adolescents and young adults in the US and worldwide (Department of Health and Human Services, 2018; Centers for Disease Control and Prevention, 2021a; Hanafin et al., 2021; Tarasenko et al., 2021). The vast majority of vape e-liquids contain nicotine – a highly addictive chemical that leads to nicotine addiction and may interfere with brain development (Department of Health and Human Services, 2016; Prochaska et al., 2022). E-cigarette aerosols also contain organic volatile compounds, metals, and chemicals that may cause respiratory harm and other health problems (Caporale et al., 2019; Hiler et al., 2021; National Academies of Sciences, Engineering, and Medicine, 2018).

Against a backdrop of rising vaping among adolescents and young adults, the 2016 report by the US Surgeon General emphasized the importance of communication to inform adolescents and young adults of the potential health harms related to vaping (Department of Health and Human Services, 2016). Following the report, several public campaigns were launched to educate youth about vaping. For instance, the Food and Drug Administration (FDA) used its *The Real Cost* tobacco prevention campaign to educate adolescents about the risks of vaping (Food and Drug Administration, 2021). The Truth Initiative, America's largest nonprofit public health group, launched the "Safer ≠ Safe" campaign to provide youth with the facts about vaping (Truth Initiative, 2019).

Evidence from smoking prevention campaigns suggests that anti-smoking campaigns have been effective in decreasing smoking (Allen et al., 2015; Wakefield et al., 2010). Anti-smoking campaign messages have changed knowledge, beliefs, and attitudes about cigarettes, which, in turn, have decreased cigarette smoking intentions and smoking behaviors among youth (Durkin et al., 2022; Hammond et al., 2012; Rath et al., 2021). The success of anti-smoking campaigns provides a basis for the notion that vaping prevention campaigns could potentially help reduce youth vaping. However, smoking prevention and vaping prevention messages differ in important ways. For instance, vaping has been touted as a less harmful alternative to smoking cigarettes, which puts vaping in a unique public health context compared to cigarette smoking (Janmohamed et al., 2022). In addition, smoking prevention messages have been communicated to the public for decades and the health effects of smoking are well-documented. In contrast, vaping devices are much newer, some of the health effects of vaping are contested, and the long-term health effects of vaping remain unknown (Sapru et al., 2020). To date, the efficacy of vaping prevention messages has not been fully evaluated.

Do vaping prevention messages have effects?

A critical question is whether vaping prevention communication efforts will have any impact at all, a key question to consider for

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several reasons. First, some past studies have revealed null effects of substance use prevention campaigns among youth. For instance, evaluations of one of the largest federally-funded substance abuse prevention campaigns – the *National Youth Anti-Drug Media* campaign – demonstrated that the campaign had no desirable effects on drug use perceptions or behaviors (Hornik et al., 2008). In the domain of vaping, Mays et al. (2016) found no significant differences in harm and addiction perceptions of e-cigarettes or thoughts about not using e-cigarettes in a test of e-cigarette ads with warnings compared to e-cigarette ads without warnings. Second, some recent studies have observed very limited effects of well-funded tobacco prevention campaigns on targeted beliefs. For example, evaluations of *This Free Life* campaign targeting lesbian, gay, bisexual, and transgender (LGBT) young adults showed the campaign had a positive impact on beliefs related to social aspects of smoking, but not on other campaign beliefs, such as beliefs on negative health risks of occasional smoking (Crankshaw et al., 2022). The *Fresh Empire* campaign – which used hip hop peer crowd targeting – was similarly found to be associated with a very limited number of campaign-targeted beliefs (Guillory et al., 2022). Third, studies have shown that prevention messages may be capable of producing boomerang effects. For instance, Wackowski et al. (2019) found that vaping warnings with a relative harm statement led to a lower recall of specific warning keywords compared to standard vaping warnings without a relative harm statement. Moreover, Boynton et al. (2022) found that vaping prevention ads with a flavor theme were rated higher on vaping appeal relative to other ads, raising concerns that flavor messages about health harms could unintentionally remind youth about the appealing aspects of vaping.

Another key question is whether vaping prevention messages can increase risk beliefs given the health and addiction focus of many vaping prevention campaigns (e.g., *The Real Cost*, Safer ≠ Safe; Kresovich et al., 2021). This is not a given, as findings from randomized controlled trials and meta-analyses consistently show *no* increases in risk beliefs about smoking after exposure to pictorial cigarette pack warnings (Brewer, Parada, et al., 2019; Noar, Rohde, et al., 2020). Thus far, studies of the effects of vaping prevention messages on risk beliefs have shown inconsistent findings. While some research has found vaping prevention messages to have no impact on risk perceptions (Majmundar et al., 2020; Wackowski et al., 2019), other studies have shown such messages to increase risk perceptions about vaping (Ariel, 2017; Katz et al., 2020). This is further complicated by the many different types of risk perceptions, including general harm (and addiction) perceptions, perceived likelihood of harm (and addiction), perceived severity of harm (and addiction), and perceived relative harm and addiction (i.e., perceptions of the harms and addiction of vaping compared to smoking (Kaufman, Persoskie, et al., 2020; Kaufman, Twesten, et al., 2020). To date, we do not know if vaping prevention messages change all, some, or none of these differing types of risk perceptions.

Theoretical frameworks informing message effects

Whether vaping prevention messages produce effects is a key theoretical and practical question. We use the Message Impact Framework (MIF) to organize and discuss the process of message effects. This framework describes several categories of antecedents of behavior change, namely attention and recall, message reactions, attitudes and beliefs, social interactions, intentions, and behavior (Noar et al., 2016). Here, we focus on three key categories of the framework most relevant to vaping prevention message effects: message reactions, risk beliefs, and intentions.

Message reactions

Message reactions deal with individuals' physiological, affective, and cognitive reactions elicited by a message, such as cognitive elaboration and negative affect (Brewer, Jeong, et al., 2019; Morgan et al., 2018). According to the Elaboration Likelihood Model (ELM; Petty & Cacioppo, 1986), individuals engage in central (involving high cognitive elaboration) or peripheral (involving low cognitive elaboration) routes in processing messages. Cognitive elaboration concerns the extent to which message recipients think about the information contained in a message (Chen et al., 2018). When individuals are motivated and have the ability to carefully process information, they are more likely to engage in central route processing; otherwise, they are more likely to engage in peripheral route processing (Petty & Cacioppo, 1986). Greater cognitive elaboration through the central route elicits attitude change that is persistent and influential in guiding behaviors, whereas low cognitive elaboration via the peripheral route is more likely to form an attitude that is easier to change and is a poor predictor of behavior change (Cacioppo et al., 1986). Meta-analyses of the effects of pictorial cigarette pack warnings have found pictorial warnings to significantly increase cognitive elaboration (Noar et al., 2016; Noar, Rohde, et al., 2020).

Since vaping prevention messages often communicate health harms or nicotine addiction risks (Kresovich et al., 2021), they also hold the potential to increase negative affect on message recipients (Witte, 1992). Negative affect refers to the experience of a negative internal feeling state, for instance, fear or anxiety (Watson & Clark, 1984), which has been found to elicit adaptive health behavior (J. Dillard & Peck, 2006; Tannenbaum et al., 2015). Meta-analyses of the effects of pictorial cigarette pack warnings have consistently found pictorial warnings to increase negative affect (Noar et al., 2016; Noar, Rohde, et al., 2020); research has shown that e-cigarette warnings may evoke more fear than no warning control condition (Andrews et al., 2019).

Risk beliefs

As per behavior change theories such as the Health Belief Model (HBM; Rosenstock, 1974), the Theory of Reasoned Action (TRA; Ajzen & Fishbein, 1980), and the Extended Parallel Process Model (EPPM; Witte, 1992), risk beliefs about a focal behavior are central to behavior change. Risk beliefs are people's subjective judgments about the severity and likelihood of harm, which may motivate engaging in a given behavior (Brewer et al., 2004). For instance, the EPPM posits that people are more likely to engage in recommended behaviors when they perceive a high threat and have high self and response efficacy (Witte, 1992). From this perspective, for a prevention message to be effective, it should increase risk beliefs in order to motivate engagement in the focal behavior (or in the case of vaping, discouraging people from vaping). Vaping poses risks of both addiction and health harms, and vaping prevention messages may differentially impact different types of risk beliefs, such as addiction perceptions, health harm perceptions, relative health harm or addiction perceptions of vaping relative to smoking cigarettes (Lazard, 2021; Mays et al., 2016; Underwood & Yang, 2018). Moreover, risk perceptions have been conceptualized and assessed differently in extant studies. For instance, while some studies measured *perceived likelihood of harm* from vaping (England et al., 2021; Lazard, 2021), others measured *perceived severity* of vaping (England et al., 2021; Underwood & Yang, 2018).

Numerous studies have examined the impact of health messages on risk beliefs (A. J. Dillard et al., 2018; Paek et al., 2016; Wackowski et al., 2019). For instance, anti-smoking ads have increased youth beliefs about the health harms of smoking (Zhao et al., 2016). Along similar lines, meta-analyses of fear appeals have found that effective fear appeals lead to higher risk perceptions (Tannenbaum et al., 2015; Witte & Allen, 2000). However, experimental studies of pictorial cigarette pack warnings have shown no impact on risk beliefs about smoking (Brewer, Parada, et al., 2019; Noar, Rohde, et al., 2020).

Intentions

Behavioral intentions refer to the readiness of an individual to perform a given behavior (Ajzen, 2002). Intentions are often used as a proxy for actual behaviors in communication research and they reliably predict behavior change (Webb & Sheeran, 2006). Many studies have examined the effects of health or risk communication messages on message receivers' intentions to take recommended actions (Ma & Miller, 2021; Nan et al., 2019). For instance, pictorial cigarette pack warnings increased intentions to not smoke among nonsmokers and intentions to quit smoking among smokers (Noar et al., 2016).

Despite the many vaping prevention campaigns used in practice (Kresovich et al., 2021), and the growing literature examining vaping prevention messages, we know little about whether they work as expected in discouraging vaping. We also do not know if they increase extant risk beliefs about vaping. Thus, a synthesis of the literature on vaping prevention messages is greatly needed. A meta-analysis can address the question of whether there is a reliable effect of vaping prevention messages on extant outcomes (Kresovich & Noar, 2020; Noar, 2006). In the current study, we synthesized the extant literature and conducted a meta-analysis of experimental studies of vaping prevention messages among adolescents and young adults. We addressed the following research question: Do vaping prevention messages impact adolescents and young adults?

Method

Search strategy

We conducted comprehensive searches of five electronic databases in October 2021: PubMed, PsycInfo, Scopus, Global Health, and Communication & Mass Media Complete. We used the following Boolean terms to search for key phrases in titles and abstracts: (adolescen* OR teen* OR "young adult*" OR youth*) AND (media OR messag* OR campaign* OR advertis* OR educat* OR Prevent* OR "text messaging" OR "texting" OR communicat* OR PSA OR "public service announcement" OR warn* OR "counter marketing" OR counter-marketing OR counter advertis* OR counter-advertis*) AND (vap* OR "electronic nicotine delivery system*" OR e-cig* OR ecig* OR juul"). A research librarian assisted with the selection of databases and the development of search terms.

Inclusion criteria

We considered for inclusion published studies that came up in our searches as well as gray literature (i.e., doctoral dissertations, theses, and conference papers) that were written in English. Further, we examined the reference lists of all included studies identified in our searches. To be eligible for inclusion, a study had to 1) be an experimental study[1], 2) have a between-subjects comparison of at least one vaping prevention message condition of any format, such as a video, text messaging, or warning label, and one control condition, 3) focus on vaping prevention (not cessation), and 4) have adolescents or young adults as the study sample (mean age does not exceed 25 years[2]).

Several studies were excluded due to their failure to meet the inclusion criteria. For instance, Liu and Yang (2020) examined how message framing (gain vs. loss) and message format (narrative vs. nonnarrative) impact responses to vaping prevention messages targeting young adults. This study was excluded because the content (i.e., vaping consequences) was the same across messages. In another case, Clayton et al. (2020) study was excluded because the treatment messages focused on encouraging current vapers to quit vaping – i.e., an explicit cessation (versus prevention) message.

The Preferred Reporting Items for Systematic Reviews and Meta-analyses (PRISMA) flow diagram in Figure 1 describes the flow of study screening and inclusion. The initial search generated 4,451 references. After removing 2,295 duplicates, there were 2,156 references. Using Covidence, a screening and data extraction tool for conducting research review, two independent coders screened the title and abstract of each reference for relevance, reducing the number to 71 for full-text retrieval. Then the two coders independently reviewed the full text of the 71 articles and noted reasons for study exclusion (see Figure 1). This full-text review resulted in 59 articles being excluded based on our inclusion criteria. During the process, if the two coders made different judgments about a particular reference, they consulted with

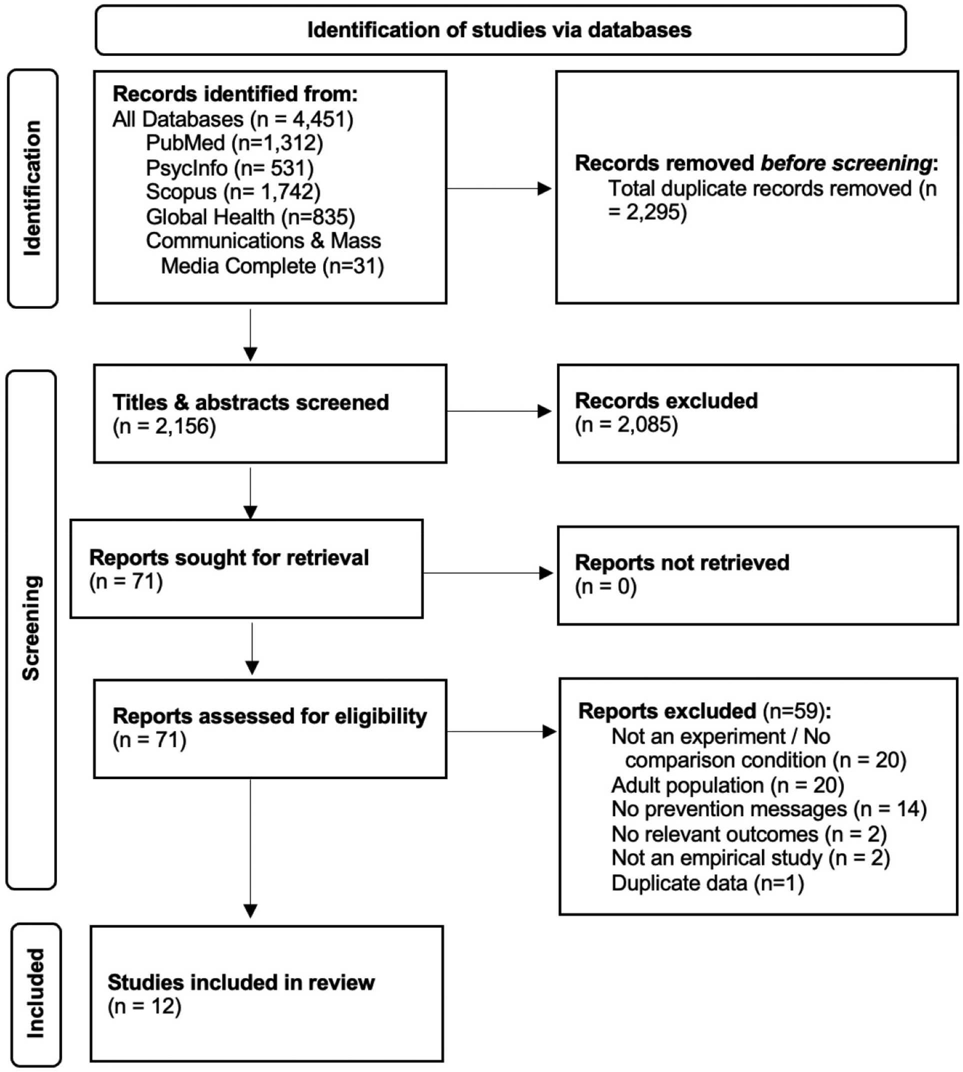

Figure 1. Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) flow diagram showing the study screening process.

a third coder to resolve the discrepancy. This process resulted in a final set of 12 articles being included in the meta-analysis.

Coding study characteristics

Using Covidence, two coders independently coded the 12 studies on sample characteristics (e.g., age, e-cigarette user status, race/ethnicity), study characteristics (e.g., study design, control message type, use of theory), and vaping prevention message characteristics (e.g., message content and format, number of message exposures, number of messages viewed). We calculated Cohen's *Kappa* for intercoder reliability, which corrects for successful coding by chance (k = 0.73). Discrepancies were resolved through discussion with a third coder. Table 1 and Supplemental Table S1 provide detail on study characteristics.

Coding outcome variables

We categorized outcome variables measured in individual studies using the Message Impact Framework (Noar et al., 2016). The framework describes a series of categories for how messages have an impact: attention and recall, message reactions, social interactions, attitudes and beliefs, intentions, and behavior. Attention and recall focus on participants' attention to vaping prevention messages and their ability to recall or recognize these prevention messages; warning reactions deal with participants' physiological, affective, and cognitive reactions toward the warning messages; social interactions refer to interpersonal communication about warning messages; attitudes and beliefs measure participants' e-cigarette or vaping-related attitudes and beliefs; intentions assess participants' willingness or readiness to use e-cigarettes; and behavior is the final category, representing the ultimate goal of prevention messages. We also included perceived message effectiveness (PME) of prevention messages, including message perceptions (i.e., evaluations of whether a message leads to further processing that promotes persuasion) and effects perceptions (i.e., a message's potential to influence intentions or behavior).

It should be noted that risk belief is an involute construct that involves many aspects (Kaufman, Persoskie, et al., 2020; Kaufman, Twesten, et al., 2020), such as the perceived seriousness of a disease (severity perceptions) and perceived likelihood of developing a disease (likelihood or susceptibility

Table 1. Participant and study characteristics of studies in the meta-analysis (n=12).

Variable	n[1]	%
Age Group		
Adolescents only	5	42
Young adults only	7	58
E-cigarette Use Status[2]		
Current Users	9	75
Ever Users	9	75
Non-Users	1	8
Experimental Design		
Between subjects	8	67
Mixed design (between and within)	4	33
Prevention Message Format		
Text only	3	25
Pictorial with text	6	50
Video	3	25
Control Condition		
Pro-vape message	5	42
Anti-vape message	4	33
No vape message	3	25
Use of Theory		
No	10	83
Yes[3]	2	17
Message Content[2]		
Health Harms	11	92
Addiction	10	83
Other (e.g., unknown health risks, tobacco marketing strategies)	2	17

[1]number of studies included in the meta-analysis.
[2]Some studies reported multiple categories; % may not add up to 100.
[3]Heuristic systematic model, protection motivation theory.

perception). For instance, risk perceptions can involve the seriousness of the health consequences of vaping, and the likelihood of developing certain illnesses due to vaping. In addition, there are perceptions of the relative risks of e-cigarettes compared with cigarettes. We defined risk beliefs about vaping broadly to include participants' perceptions of vaping related to health harms or addiction, whether they were 1) harm or addiction perceptions; 2) perceived likelihood of harm or addiction; 3) severity perceptions; or 4) relative harm or addiction perceptions. To examine whether vaping prevention messages had differential effects on these different types of risk perceptions, we examined them as separate outcomes in our analyses.

Effect size extraction and calculation

Effect sizes characterizing the impact of vaping prevention messages compared to control conditions were computed using the standardized mean difference, or d, which is the difference between vaping prevention message and control condition means divided by a pooled standard deviation (Lipsey & Wilson, 2001). Because d can be upwardly biased when based on small sample sizes, we applied the recommended statistical correction for this bias (Lipsey & Wilson, 2001). In cases where a d could not be computed from the data reported in the article, data were requested from the article authors. Effect sizes were computed on all vaping-related outcomes assessed in two or more independent samples, using measures gathered after participants viewed their respective messages.

For studies that included more than one vaping prevention message condition, an effect size was calculated for the comparison of each vaping prevention message condition and the control condition, to take full advantage of the rich data included in individual studies. Thus, a study with two prevention message conditions and a control condition would contribute two effect sizes to the meta-analysis (i.e., message A compared to control, message B compared to control). This does not violate the assumption of independence given that participants in each experimental condition are independent (Harkin et al., 2016; Webb & Sheeran, 2006). In these cases, the sample size for the control group was divided by the number of prevention message conditions to avoid double counting participants when weighting effect sizes (Higgins & Greens, 2011).

Meta-analytic approach

Effect sizes were weighted using inverse variance weights and combined using random-effects meta-analytic procedures (Lipsey & Wilson, 2001). The Q and I^2 statistics were calculated to examine the relative amount of variability among the effect sizes relative to that expected on the basis of sampling variation (i.e., relative heterogeneity). Publication bias was also examined. All analyses were conducted using the Comprehensive Meta-Analysis software Version 2.2.046.

Results

Study characteristics

The 12 studies had a cumulative $N = 6{,}622$, adolescents and young adults. All studies were conducted in the United States. These studies took place over a 5-year time period (2016–2021). Study samples consisted of mostly (7 studies, 58%)

young adults, with 5 studies (5 studies, 42%) of adolescents. Studies applied either fully between-subjects (8 studies, 67%) or mixed (4 studies, 33%) designs. Most studies (83%) did not use theory, and experiments varied in their control conditions (Table 1). See Supplementary Table S1 for detailed information about individual studies.

We grouped all vaping-related outcomes assessed in the 12 studies into seven categories: attention and recall, message reactions, social interactions, knowledge, attitudes and beliefs, intentions, behavior, and PME. Across these studies, a total of 35 unique vaping-related outcomes were assessed (Figure 2). Fourteen outcomes were assessed in two or more independent

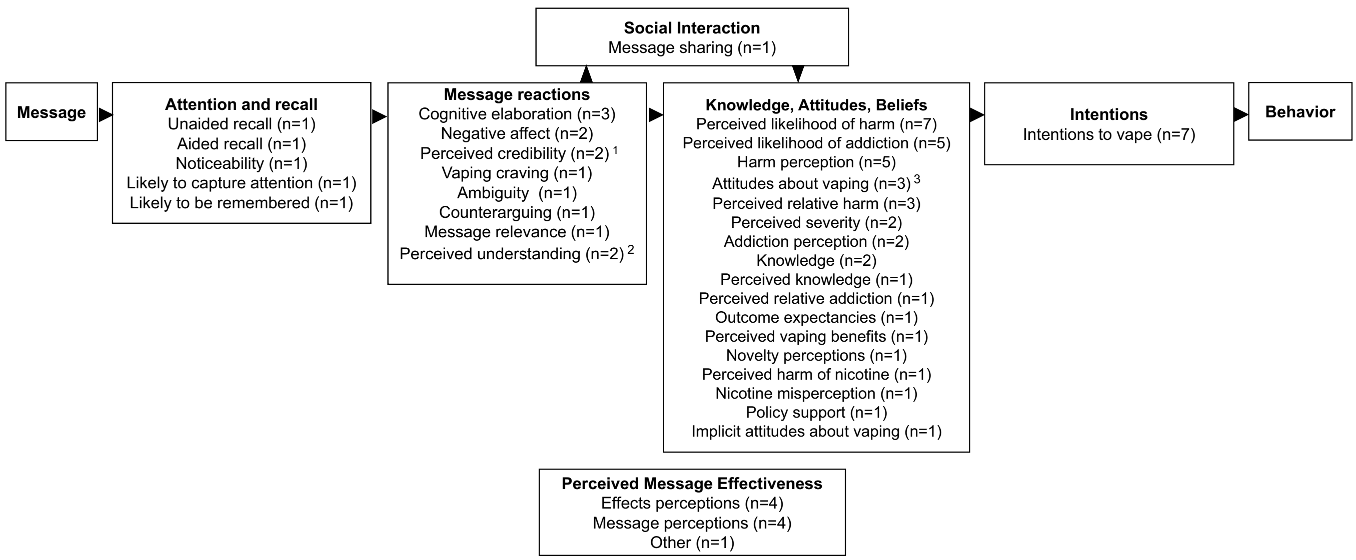

Figure 2. Outcomes assessed in the 12 vaping prevention message studies (*n* = number of studies).
[1]Two studies assessed "Perceived credibility", but this outcome was not meta-analyzed due to lack of data for a control condition.
[2]Two studies assessed "Perceived understanding" but this outcome was not meta-analyzed due to lack of data for a control condition.
[3]Three studies assessed "Attitudes about vaping", but this outcome was not meta-analyzed due to lack of data in one study and lack of data for a control condition in another study.

Table 2. Outcomes examined in the meta-analysis.

Outcome Category and Construct	Definition	Example Item
Message Reactions		
Cognitive elaboration	Thinking about the message's content, such as the harms of vaping	"How much does the message make you think about the harmful effects of vaping?" (Lazard, 2021)
Negative affect	Negative emotional reactions to the message, such as fear or disgust	"Did this information make you feel frightened?" (Underwood & Yang, 2018)
Craving for vaping	Experiencing an urge to vape	"After having viewed the e-cigarette ad, how much do you want to smoke an e-cigarette right now?" (Andrews et al., 2019)
Beliefs		
Harm perceptions	Beliefs about potential harms caused by e-cigarette use	"How much do you think people harm themselves when they use electronic cigarettes (e-cigarettes)?" (Katz et al., 2020)
Perceived likelihood of harm	Beliefs that e-cigarette use is likely to lead to vaping-related disease	"If I vape regularly, I will hurt my health." (Lazard, 2021)
Perceived relative harm	Beliefs that e-cigarette use causes more or less harm than other substance use	"Would you say electronic cigarettes are less or more harmful to one's health than smoking regular cigarettes?" (Mays et al., 2016)
Perceived severity	Beliefs that the health harms of vaping-related disease are serious	"I believe that the harmful effects of e-cigarette use are severe." (Underwood & Yang, 2018)
Addiction perceptions	Beliefs about potential addiction caused by e-cigarette use	"Using e-cigarettes can be addictive." (Andrews et al., 2019)
Perceived likelihood of addiction	Beliefs that e-cigarette use is likely to lead to vaping-related addiction	"If I vape, I will become addicted to vaping." (Rohde et al., 2021)
Perceived relative addiction	Beliefs that e-cigarette use is more or less addictive than other substance use	"Would you say e-cigarettes are less or more addictive than smoking regular cigarettes?" (Mays et al., 2016)
Vaping knowledge	Participant's knowledge of e-cigarette use consequences	"Flavoring chemicals found in vaping devices have been linked to a serious lung disease." (England et al., 2021)
Intentions		
Intentions to vape	A person's likelihood of vaping	"How likely is it that you will vape soon?" (Noar, Rohde, et al., 2020)
Perceived Effectiveness		
Message perceptions	Judgements of whether a message will lead to further processing that promotes persuasion	"The ads 1) grab my attention, 2) are informative . . . " (Rohde et al., 2021)
Effects perceptions	Judgments of a message's potential to change behavioral antecedents or behavior itself	"The ads make vaping seem like a bad idea to me" (Noar, Rohde, et al., 2020)

samples and were meta-analyzed. See Table 2 for the 14 outcomes, definitions, and example items.

Effects of vaping prevention messages on message reactions

Vaping prevention messages had no significant impact on message reactions (See Table 3). Specifically, compared to control, vaping prevention messages did not affect cognitive elaboration ($d = 0.47$, $p = .178$), negative affect ($d = 0.17$, $p = .227$), or craving for vaping ($d = -0.21$, $p = .071$). Heterogeneity statistics showed that cognitive elaboration ($I^2 = 92.31$) and negative affect ($I^2 = 50.84$) exhibited relative heterogeneity, while craving for vaping was homogenous.

Effects of vaping prevention messages on risk beliefs and behavioral intentions

Vaping prevention messages affected vaping risk beliefs and behavioral intentions. Compared to control, vaping prevention messages increased several harm risk beliefs: harm perceptions ($d = 0.30$, $p < .001$), perceived likelihood of harm ($d = 0.23$, $p < .001$), and perceived relative harm ($d = 0.14$, $p = .036$). Vaping prevention messages also increased several addiction risk beliefs: addiction perceptions ($d = 0.39$, $p < .001$), perceived likelihood of addiction ($d = 0.22$, $p < .001$), and perceived relative addiction ($d = 0.33$, $p = .015$), relative to control. The impact of vaping prevention messages (vs. control) on perceived severity was not significant ($d = 0.17$, $p = .084$). Only harm perceptions ($I^2 = 45.08$) and perceived relative addiction ($I^2 = 44.48$) showed relative heterogeneity (See Table 3).

Vaping prevention messages also significantly increased vaping knowledge ($d = 0.37$, $p < .001$) and decreased vaping intentions ($d = -0.09$, $p = .022$), compared to control. Heterogeneity statistics revealed relative heterogeneity for vaping knowledge ($I^2 = 55.61$). Forest plots of harm and addiction risk perceptions and intentions to vape are displayed in Figures 3, 4, and 5.

Effects of vaping prevention messages on perceived message effectiveness (PME)

Vaping prevention messages had a significant impact on PME. Messages increased both message perceptions ($d = 0.57$, $p < .001$) and effects perceptions ($d = 0.55$, $p < .001$), compared to control, with both effect sizes were relatively heterogeneous ($I^2 = 66.02$ and $I^2 = 86.75$, respectively; see Table 3).

Influence of control condition on effect size estimates

Given that studies varied in their control conditions, including no, anti-, and pro-vape messages as controls, we examined whether control condition influenced effect size estimates. We first looked at the distribution of control conditions for all outcomes with 10 or more effect sizes. For harm perceptions, all studies but one used the same type of control condition, and thus an analysis of that outcome was not possible. For perceived likelihood of harm, perceived likelihood of addiction, and intentions to vape, we examined control condition type (no, anti-, or pro-vape message) as a moderator of effect size. Results indicated no significant difference among effect sizes across these outcomes based on control condition type, as indicated by the non-significant Q_B test (Table 4). This suggested that type of control condition did not meaningfully influence effect size estimates.

Publication bias

We also examined publication bias on each meta-analyzed outcome using funnel plots and the trim-and-fill method (Shi & Lin, 2019). The trim-and-fill suggested that no publication bias existed on any of the effect size distributions except for the vaping knowledge and message perceptions. After adjusting for publication bias, the effect size for vaping knowledge slightly decreased from $d = 0.37$ to $d = 0.32$, and the effect size for message perceptions decreased from $d = 0.57$ to $d = 0.40$.

Discussion

Vaping among adolescents and young adults is alarmingly high, which imposes significant addiction and other health

Table 3. Effectiveness of vaping prevention messages: mean weighted effect sizes (*d*) and heterogeneity statistics.

	N	*k*	*d*	95% CI	*p*	Q	p_Q	I^2
Message reactions								
Cognitive elaboration	643	5	0.47	[−0.214, 1.151]	.178	52.02	<.001	92.31
Negative affect	547	5	0.17	[−0.108, 0.454]	.227	8.14	.087	50.84
Craving for vaping	340	2	−0.21	[−0.432, 0.018]	.071	0.01	.904	0
Attitudes and beliefs								
Harm perceptions	1,574	11	0.30	[0.143, 0.447]	<.001	18.21	.052	45.08
Perceived likelihood of harm	3,100	13	0.23	[0.155, 0.307]	<.001	5.83	.924	0
Perceived relative harm	1,139	8	0.14	[0.009, 0.268]	.036	2.04	.958	0
Perceived severity	475	4	0.17	[−0.023, 0.363]	.084	2.30	.512	0
Addiction perceptions	776	4	0.39	[0.24, 0.541]	<.001	2.54	.468	0
Perceived likelihood of addiction	2,826	10	0.22	[0.138, 0.295]	<.001	3.79	.924	0
Perceived relative addiction	436	2	0.33	[0.064, 0.603]	.015	1.801	.18	44.48
Vaping Knowledge	1,197	4	0.37	[0.177, 0.555]	<.001	6.76	.08	55.61
Intentions								
Intentions to vape	2,786	16	−0.09	[−0.163, −0.012]	.022	11.93	.684	0
Perceived message effectiveness (PME)								
Message perceptions	1,204	3	0.57	[0.35, 0.80]	<.001	5.89	.053	66.02
Effects perceptions	1,976	6	0.55	[0.257, 0.84]	<.001	37.73	.000	86.75

N=number of participants; *k*=number of effect sizes; *d*=standardized mean difference (pooled effect size); p_Q=p value associated with Q statistic.

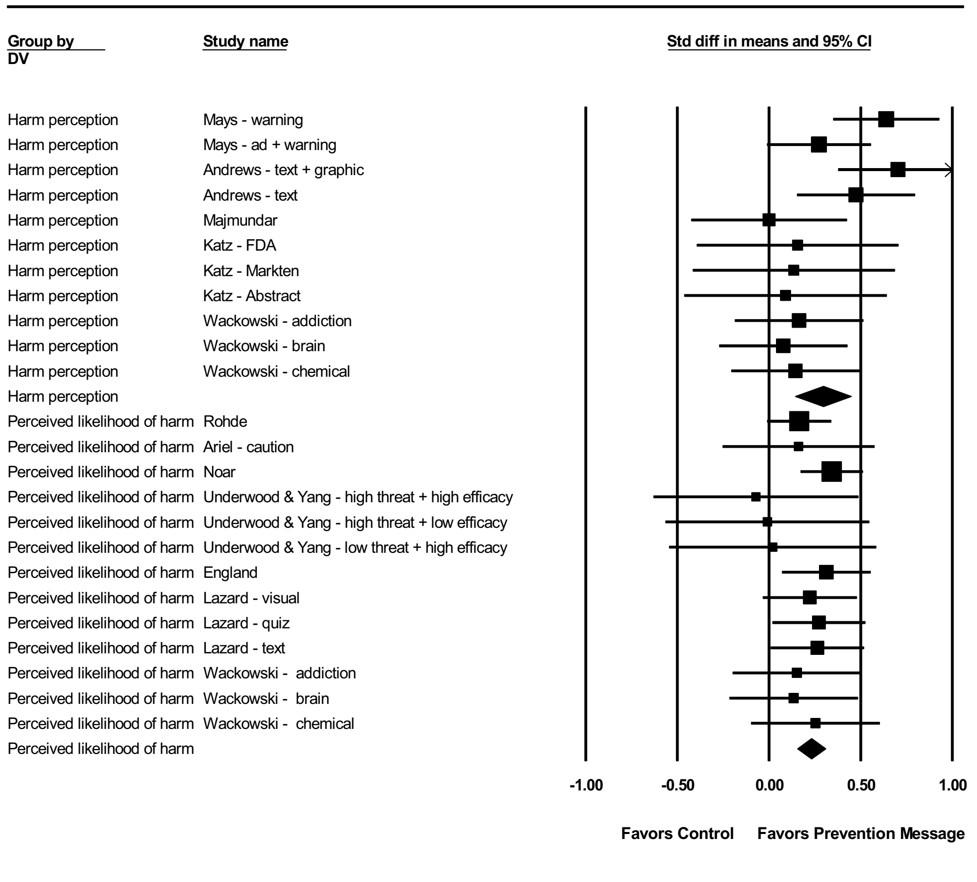

Figure 3. Forest plots of effect sizes and 95% confidence intervals for harm perceptions and perceived likelihood of harm.

risks on users (Hanafin et al., 2021; Prochaska et al., 2022). Vaping prevention messages are widely used to communicate the risks of e-cigarette use and discourage vaping (Food and Drug Administration, 2021). However, the effects of these prevention messages have been unknown. In the current study, we conducted the first meta-analysis of experimental studies on vaping prevention messages to examine whether these messages have their intended effects. A meta-analysis of 12 studies including 6,622 adolescents and young adults revealed clear effects. Compared to control, vaping prevention messages increased six out of seven types of risk beliefs about vaping, vaping knowledge, and PME, and to a lesser extent, decreased intentions to vape among adolescents and young adults. Findings of this meta-analysis suggest that vaping prevention messages hold potential to discourage vaping among adolescents and young adults.

The vaping prevention messages identified in our search tended to focus on the health harms or addiction of vaping (Sanders-Jackson et al., 2019; Wackowski et al., 2019). Compared to anti-smoking messages which have used a wide variety of themes (e.g., combatting the social norm of smoking or industry manipulation; Allen et al., 2015), the vaping prevention messages in this meta-analysis tended to focus on the addiction and health consequences of vaping, with only one study including messages about the marketing strategies of tobacco companies (England et al., 2021). Prior research examining the types of vaping prevention messages used by practitioners has found most messages to focus on addiction and other health harms of vaping (Kresovich et al., 2021), suggesting that the message themes in our meta-analysis are similar to those used by public health practitioners.

Vaping prevention messages examined in this meta-analysis were delivered primarily through text-only messages, text plus imagery, or videos (Rohde et al., 2021; Sontag et al., 2019), with some messages being presented in formats that mimic social media sites (e.g., Lazard, 2021). In terms of outcome measures, the studies in this meta-analysis included a wide variety of constructs. While some studies focused on vaping beliefs such as risk perceptions and intentions to vape (England et al., 2021; Underwood & Yang, 2018), others also examined message-related responses, such as PME and negative affect (Andrews et al., 2019; Noar, Rohde, et al., 2020). The variety of message effects measures suggests that scholars are taking different aspects into consideration when examining the impact of vaping prevention messages, which may help the field garner a more comprehensive understanding of how vaping prevention messages operate.

Importantly, compared to control, vaping prevention messages decreased vaping intentions among adolescents and young adults, but this effect was very small. This finding is somewhat heartening since behavior change is the gold standard of successful messages (Snyder & LaCroix, 2001) and behavioral intentions are the best predictor of behavior change (Eccles et al., 2006; Webb & Sheeran, 2006). However, this very small effect may be the result of vaping intentions being more

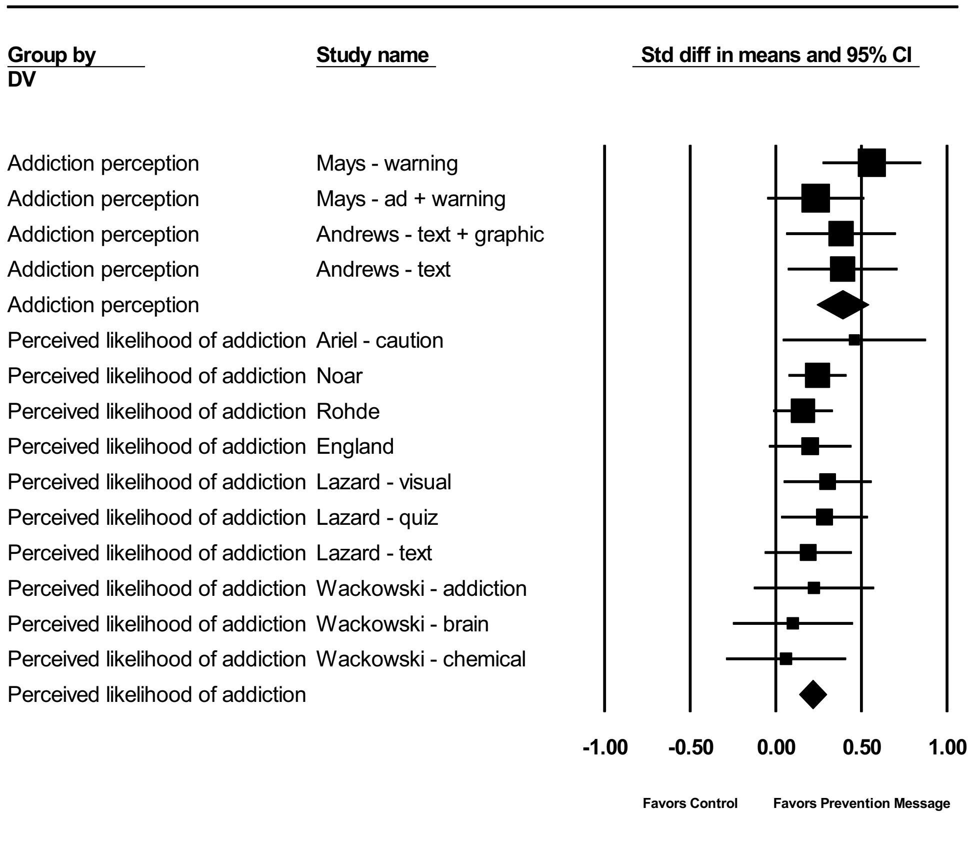

Figure 4. Forest plots of effect sizes and 95% confidence intervals for addiction perceptions and perceived likelihood of addiction.

difficult to change than risk beliefs about vaping. In most studies, outcomes were assessed after just a single exposure to messages, and larger effects may occur in real-world communication campaigns with multiple exposures over time. This should be the subject of additional research, as well as understanding how young people process vaping prevention messages in the real world (e.g., on social media).

Vaping prevention messages also impacted a series of risk beliefs about both harm and addiction. We categorized and meta-analyzed several different types of risk beliefs, finding that vaping prevention messages increased a series of six (out of seven) risk beliefs about vaping, including both harm and addiction beliefs, compared to control. The finding is consistent with other literature that demonstrates risk-related messages can be successful in increasing risk perceptions (A. J. Dillard et al., 2018; Gerrard et al., 1999; Tannenbaum et al., 2015), and strongly suggests that vaping prevention messages are an effective strategy to increase vaping risk beliefs. The current findings on risk beliefs are also consistent with a systematic review where nicotine addiction messages related to nicotine vaping products were associated with greater health and addiction risk perceptions (Erku et al., 2021). Furthermore, it is plausible that the effect of vaping prevention messages on decreasing vaping intention works through increasing risk beliefs as suggested by behavior change theories such as the TRA and EPPM (Fishbein & Ajzen, 2010; Witte, 1992), but future studies testing risk beliefs as mediators in influencing intentions and behavior are needed to test this empirically.

Vaping prevention messages significantly increased PME – both effects perceptions and message perceptions. The significant effect on PME enhances confidence in these prevention messages, given that PME ratings have been demonstrated to predict the actual effectiveness of health messages in the context of tobacco prevention and control (Bigsby et al., 2013; Noar, Barker, et al., 2020). Future formative research may benefit from including PME measure to select more effective candidate messages before launching campaigns. Vaping related knowledge also increased, which is important given that youth and young adults lack knowledge about the harms of vaping due to the novelty of e-cigarettes and the vast amount of pro-vaping marketing that reaches adolescents (England et al., 2021; Kong et al., 2015).

The effect of vaping prevention messages on other message reaction variables such as cognitive elaboration and negative affect was not significant, which is a bit surprising as they are key mechanisms of persuasion (Loewenstein et al., 2001; Petty & Cacioppo, 1986). However, given that we only had a modest number of studies assessing cognitive elaboration and negative affect combined with the fact that many of the prevention messages were not emotionally evocative (e.g., text-only messages), it is too early to make firm conclusions about the effects of vaping prevention messages on cognitive elaboration or negative affect. More nuanced, implicit measurement instruments, for instance, psychophysiological measures (e.g., skin conductance, facial electromyography, heart rate) may also be used to explore the effect of message reactions. Such measures

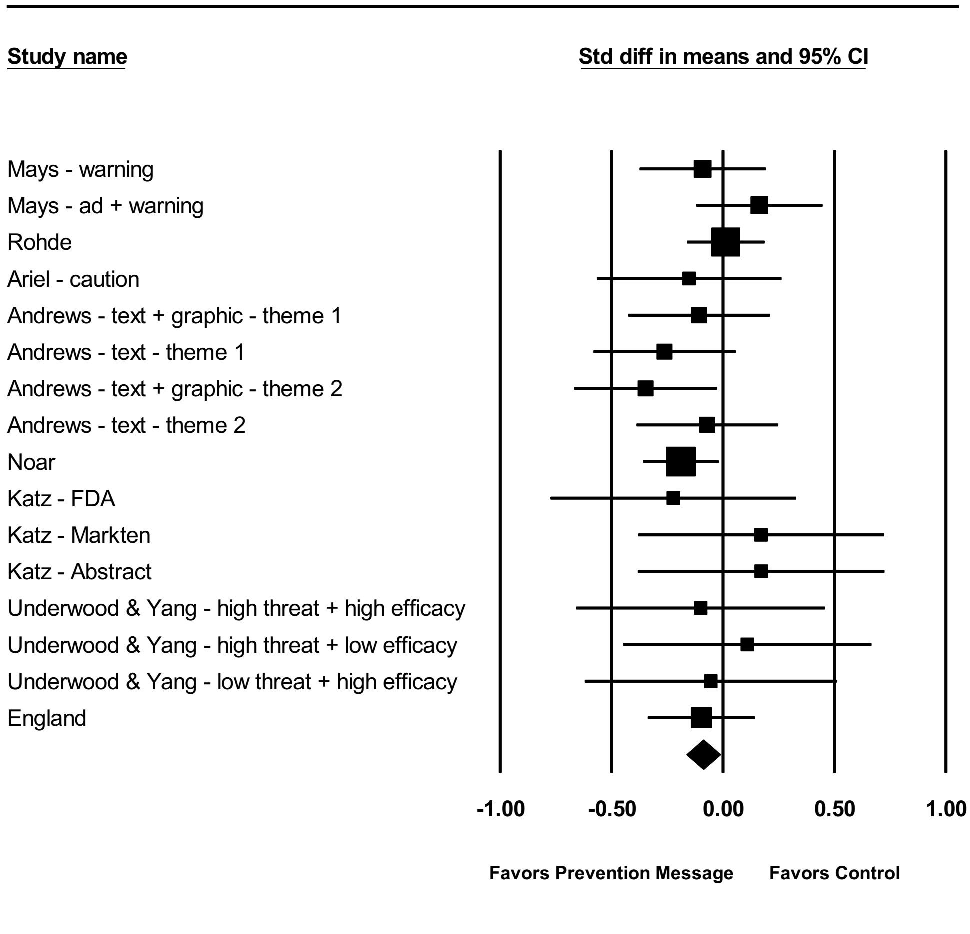

Figure 5. Forest plots of effect sizes and 95% confidence intervals for intentions to vape.

Table 4. Control conditions as potential moderators of effect size.

	k	*d*	95% CI	Q_B	Q_B *p* value
Perceived likelihood of harm				0.80	.67
Pro-vape message	4	.18	[−.006, .358]		
Anti-vape message	5	.21**	[.079, .342]		
No vape message	4	.27***	[.144, .395]		
Perceived likelihood of addiction				0.30	.86
Pro-vape message	4	.19*	[.009, .372]		
Anti-vape message	2	.20**	[.082, .324]		
No vape message	4	.24***	[.118, .368]		
Intentions to vape				2.85	.24
Pro-vape message	6	.01	[−.147, .167]		
Anti-vape message	5	−.082	[−.195, .031]		
No vape message	5	−.166*	[−.299, −.034]		

k=number of effect sizes, *d*=standardized mean difference, CI=confidence interval.

can detect nuanced responses to messages during message exposure that post-exposure self-reports may fail to capture.

Findings of the current meta-analysis suggest different theoretical mechanisms for vaping prevention messages when compared to pictorial cigarette pack warnings, with meta-analyses finding *no* impact of these warnings on cigarette smoking risk perceptions (Noar et al., 2016; Noar, Rohde, et al., 2020). Instead, pictorial warning labels increased fear and other negative emotions as well as cognitive elaboration. A possible explanation for these differences is that the harms of cigarette smoking are well-documented, and people's perceptions of the harms of cigarette smoking are likely difficult to change. In contrast, e-cigarettes are novel products whose long-term health consequences are still not known (Livingston et al., 2022; Sapru et al., 2020), leaving many youths to underestimate the harms of vaping (Centers for Disease Control and Prevention, 2021b). In addition, pictorial cigarette warning labels feature vivid visual presentations of the gruesome consequences of smoking cigarettes (e.g., diseased lungs, hole in the throat), compared to vaping prevention messages which have tended to be text-only statements or less graphic images about the addiction and other health consequences of vaping (Food and Drug Administration, 2016; Brewer, Jeong, et al., 2019). Therefore, harm beliefs about vaping should be (and appear to be) more malleable compared to harm beliefs about cigarette smoking.

The current findings appear to suggest that while the creation of vaping prevention messages more or less borrows from the successful experience of anti-smoking messages and pictorial warning labels, the theoretical mechanisms of vaping prevention messages may be somewhat different from that of pictorial warnings for cigarettes. It would seem that vaping prevention messages impact vaping intention primarily through cold cognition (i.e., changing risk beliefs about vaping) — cognitive processing of stimuli that is independent of emotional involvement (Roiser & Sahakian, 2013), whereas pictorial cigarette warning labels stimulate more hot cognition (i.e., affective and some cognitive appraisals) — motivated reasoning where individuals' thinking is colored by their emotional states (Lodge & Taber, 2005).

Conclusion

This is the first meta-analysis, to our knowledge, to examine the impact of vaping prevention messages. It proffers valuable theoretical and practical implications for the effectiveness of vaping prevention messages and public health campaigns to discourage vaping among adolescents and young adults. We found that compared to control, vaping prevention messages increased a series of both addiction and harm risk perceptions, vaping knowledge, and PME, and decreased intentions to vape. These findings support the central role of risk beliefs in the efficacy of vaping prevention messages, aligning with communication and behavior change theories (Fishbein & Ajzen, 2010; Witte, 1992). In terms of practical implications, our findings suggest the great potential of vaping prevention messages focused on the addiction and health harms of vaping in discouraging vaping among adolescents and young adults.

In a recent systematic review, Erku et al. (2021) stated that "Meta-analysis was not feasible because of study heterogeneity in terms of study design, message manipulations and outcome measurement." (p. 3293). Given that more studies have been published since Erku, we were able to identify twelve studies on which several message response variables were meta-analyzed. Still, the number of studies in our meta-analysis was modest, and additional meta-analyses should be undertaken as this literature continues to grow and new studies emerge (e.g., Patterson et al., 2022).

Given the modest literature on vaping prevention messages, an exploration of the impact of theory-guided communication strategies (e.g., gain vs. loss frame) was not possible since the number of available studies was inadequate to conduct such a meta-analysis. For example, we located only 2 studies that examined gain versus loss messages for vaping prevention (Kong et al., 2016; Liu & Yang, 2020). As more studies examine theory-based strategies in message design for vaping prevention, additional meta-analyses can be undertaken.

In conclusion, the current meta-analysis provides evidence that vaping prevention messages impact a series of addiction and health harm risk beliefs about vaping and other outcomes. This should reinvigorate efforts – including by local, state, and federal organizations – to develop and disseminate effective communications with young people about the harms of vaping to their bodies and minds in order to discourage use.

Notes

1. We only included experimental studies since experiments can isolate the effects of vaping prevention messages on outcomes and thus offer stronger evidence of the causal impact of prevention messages compared to other types of studies (e.g., survey, quasi-experiment).
2. If studies only reported age range (not mean age) whose mean age is impossible to exceed 25 years and other criteria were met, these studies were also included.

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